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COSC 5580
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Project Assignment 2 - Report

Data Preprocessing

Missing Data

Our data for Japan and the US is combined in a single csv file (us_japan_data.csv), so the first step was to analyze the amount of missing values. We found a considerable amount of missing data and hence decided to use a combination of "Regression (on attributes which had more missing data) and Mean imputations (on attributes which had fewer missing values).
Note: Based on the sum of missing values analysis which is present below with screenshots, we decided which attributes required regression.

For US, there are 2 attributes which need regression (gdp_pc_US, govt_debt_us). For Japan, only 1 attribute required regression (govt_debt_j). As it can be seen from Figure 1 and Figure 2.

```
date                      0
exchange_rate_USD_JY    536
cpi_us                    1
inflation_us            175
interest_r_us             0
gdp_pc_us              9135
govt_debt_us           9135
dtype: int64
```

Figure 1: Missing Data for US

Noise outliers

To detect noise data, we first plotted the variables (Figure), to visually check if there are sudden jumps or abrupt value changes. We have included some of the plots in the report and observed that there are very few data points showing these properties. We planned not to remove those values this early in our analysis because we will use binning and anomaly detection (using LOF) to have some concrete evidence of outliers. Additionally, we used box plots to visualize the range of few variables, Figure 3 and Figure 4.

```
ject-2-group-3/Japan_missing_data.py
date                        0
exchange_rate_USD_JY      536
cpi_j                     350
interest_r_j             1655
inflation_j               175
gdp_pc_j                  175
govt_debt_j              2781
dtype: int64
```

Figure 2: Missing Data for JP

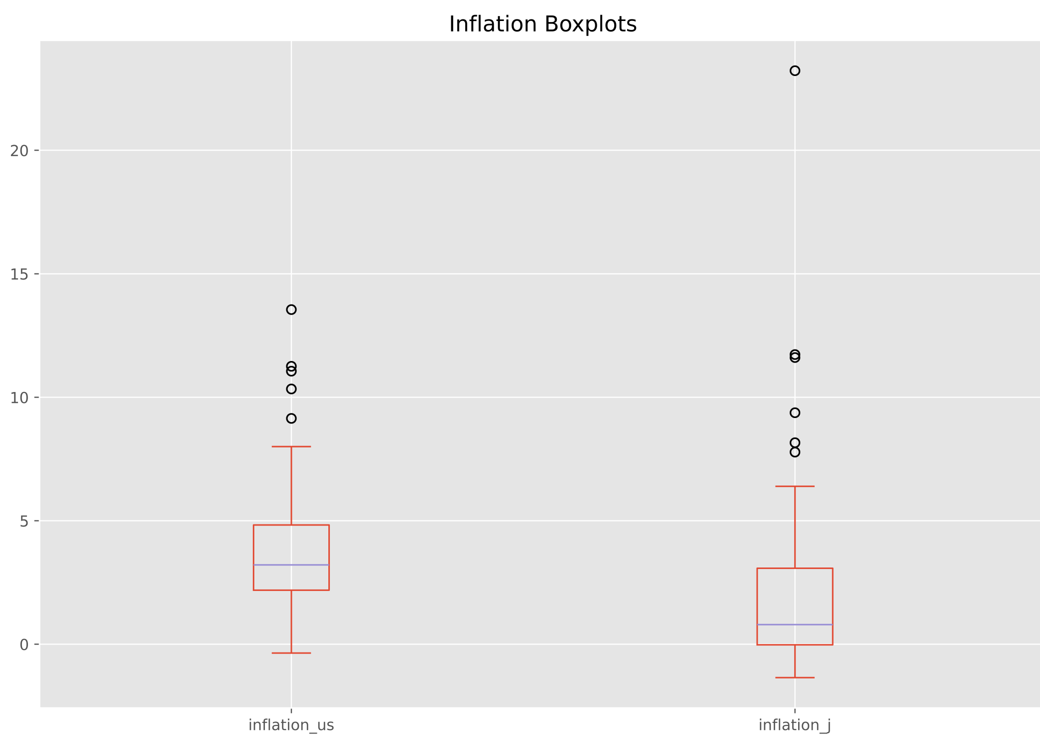


Figure 3: Inflation Boxplot

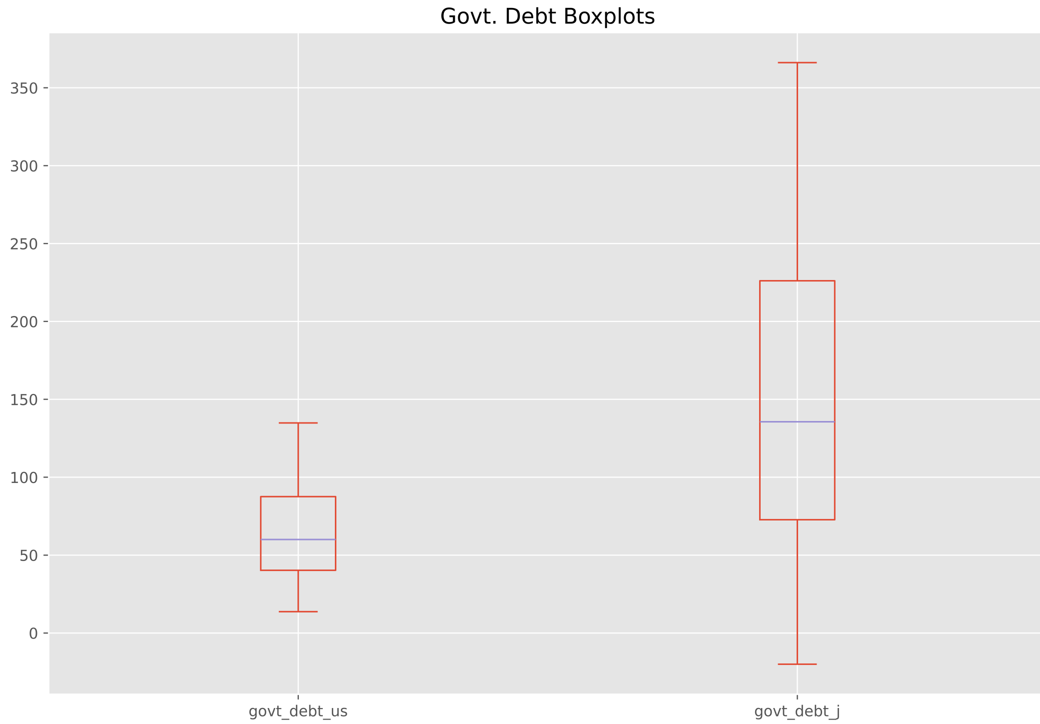


Figure 4: Govt. Debt Boxplot

Duplicates

We used the inbuilt *pandas* function **pandas.DataFrame.duplicated**, to check the pressence of duplicate values. Since our index is a datetime object we did not encounter any duplicates. No further processing here.

Basic Statistical Analysis and data cleaning insight

Statistical Features

Table 1 shows all the required statistical values for 10 attributes. It can be observed that the mean inflation of Japan is lower compared to US. This can also be seen from the fact that the mean CPI of US is much higher than that of Japan. One interesting thing to note here is that, although Japan has lower interest rates but the Debt held by the government is much higher than the US.

Anomaly Detection - LOF

For anomaly detection, we used the *Local Outlier Factor (LOF)* algorithm with three values of k. We are using the scikit-learn library and *negative_outlier_factor_* as a metric to check if the data points are inliers or outliers. We found that for using a threshold score of -0.98 works best in our case with the value of $k = 10$. We did this because we don't want to remove lot of data from our dataset as in the future we may want to perform some time-series analysis on

Table 1: Statistical Properties

Feature	**Mean**	**Median**	**Standard Deviation**
cpi_us	157.32	158.7	71.29
cpi_j	86.68	96.41	18.33
inflation_us	4.04	3.21	2.89
inflation_j	2.49	0.79	4.27
gdp_pc_us	41945.7	42114.16	10700.91
gdp_pc_j	27799.55	30636.26	6802.58
interest_r_us	4.94	4.94	3.99
interest_r_j	2.78	2.73	2.52
govt_debt_us	63.81	60.05	27.22
govt_debt_j	148.02	135.60	79.99

it. Eliminating lot of data could potentially cause problems in our analysis due to large gaps in data. We observe that the data was sufficiently clean and ready for analysis We also plotted the number of outliers along with their outlier factor scores for all the values of k.

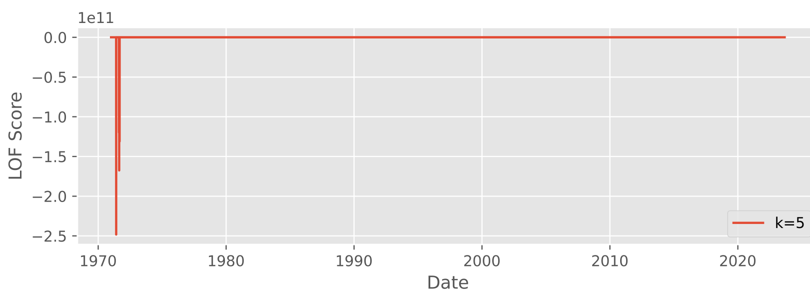


Figure 5: LOF Scores for k = 5

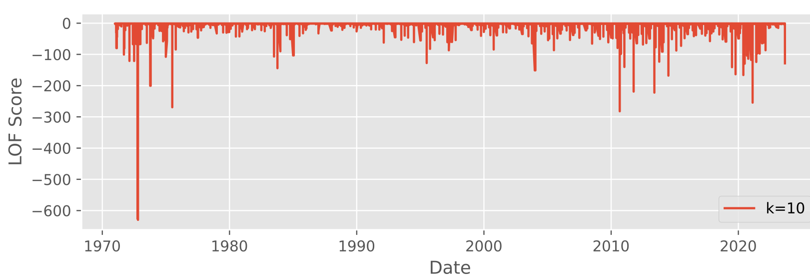


Figure 6: LOF Scores for k = 10

Binning

Binning attribute: govt_debt_j. Binning details can be found here 8

- The data is distributed among different data ranges, which shows a variety of values in the dataset

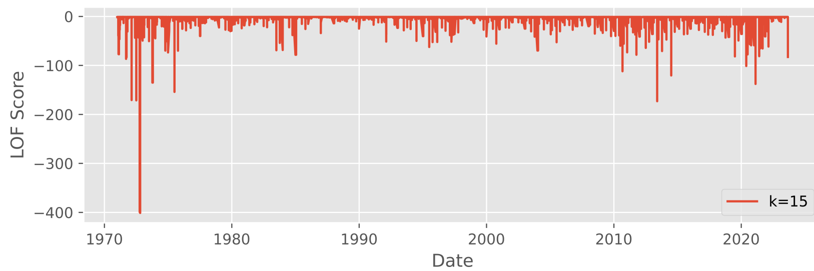
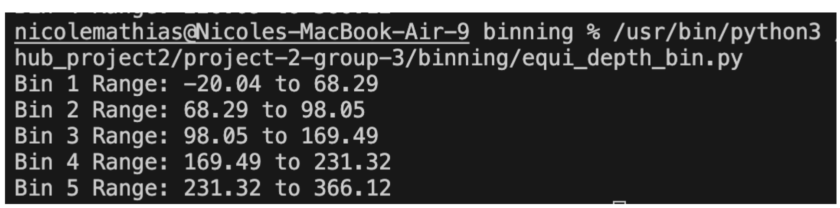


Figure 7: LOF Scores for k = 15

- The widths in every bin vary, which shows the difference in density across different data ranges.
- Since the bin ranges are not symmetric, it indicates potential data skewness in the distribution of data.
- There are some negative values in bin 1 which means there is a possibility that there are outliers or special data cases in that area.
- Since bin 4 and bins 5 have a larger range, which indicates that a significant portion of the data (government dept for Japan) has higher values especially in bin 5.



```
nicolemathias@Nicoles-MacBook-Air-9 binning % /usr/bin/python3
hub_project2/project-2-group-3/binning/equi_depth_bin.py
Bin 1 Range: -20.04 to 68.29
Bin 2 Range: 68.29 to 98.05
Bin 3 Range: 98.05 to 169.49
Bin 4 Range: 169.49 to 231.32
Bin 5 Range: 231.32 to 366.12
```

Figure 8: Binning Details

Histograms and Correlations

Histograms

Looking directly at the histograms 9 of the four attributes cannot be obvious, so we've made two more easily viewable histograms 10 and 11 for countries as a unit. In this histogram, we can see that the frequency distribution of the inflation rate is similar to that of the interest rate. It shows that there can be some correlation between the two. According to the Phillips curve, there is an inverse relationship between national inflation rates and exchange rates. This relationship occurs because central banks raise interest rates when national inflation is high to cool the economy and reduce inflationary pressures. Conversely, when inflation is low, the central bank may lower interest rates to stimulate economic activity.

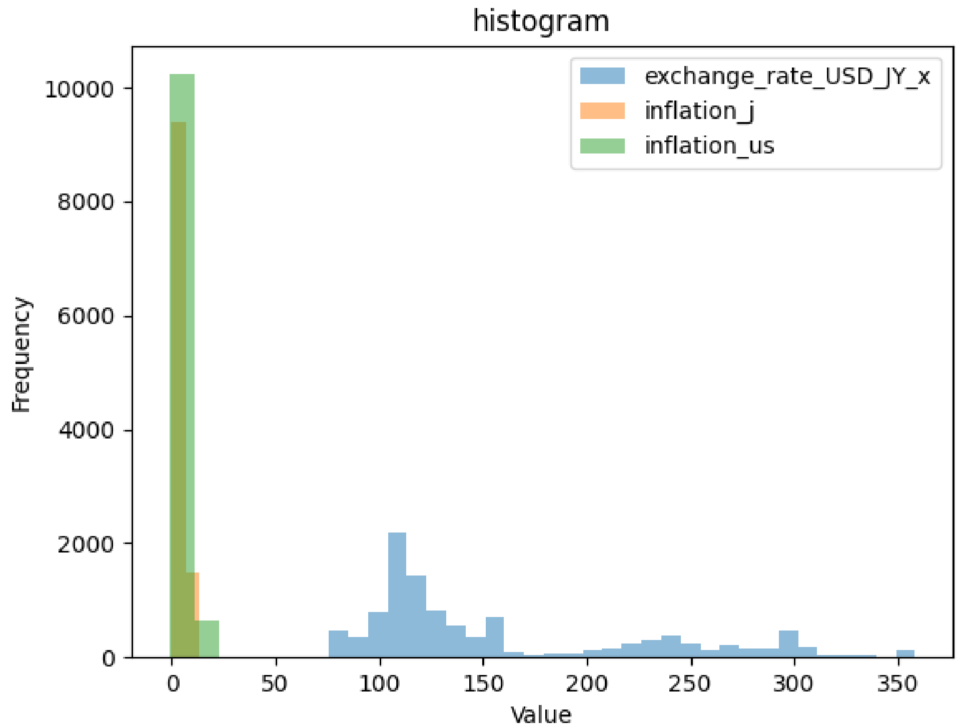


Figure 9: Histogram

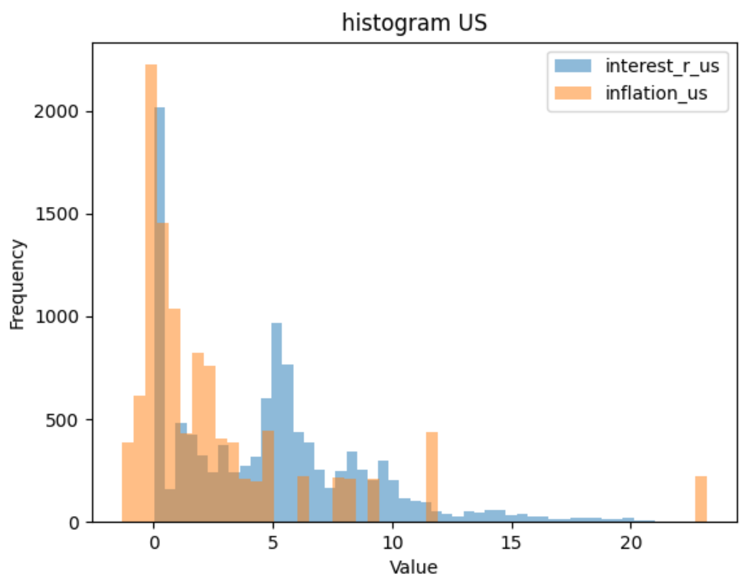


Figure 10: Histogram US

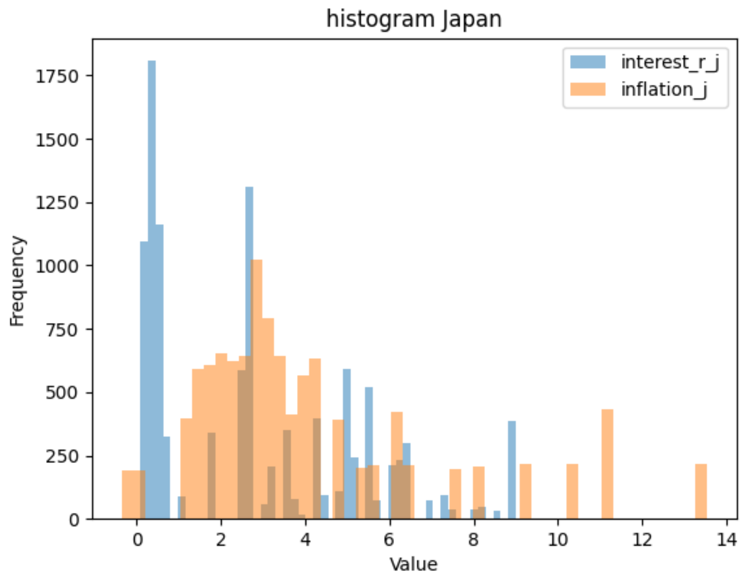


Figure 11: Histogram JP

Correlation

These three scatterplots 12, 13, 14, reveal a substantial link between exchange rates, interest rates, and purchasing power. Both of them correlate 0.8. The exchange rate-CPI link is essential for several reasons. Exchange rates affect import prices, which involves a country's CPI. Currency depreciation can raise import prices and CPI inflation, mainly if a country imports a lot. Furthermore, a country's exchange and CPI inflation rates are usually related. A country with high inflation depreciates its currency compared to one with low inflation. Higher inflation lowers a currency's purchasing power, making it less valuable on the foreign exchange market. If a country has lower inflation than its trading partners, its currency may rise because it retains its purchasing power. Overall, we also can find that international inflation rates are indirectly linked. If U.S. inflation is strong and Japanese inflation is low, trade dynamics may shift. A dollar devaluation against the yen would make U.S. exports more competitive in Japan and make Japanese imports more expensive for Americans. The relative demand for goods from both countries would affect the trade balance and possibly inflation.

Cluster Analysis

For cluster analysis we used the *Silhouette score* as a metric to access the quality of clusters.

K-Means

Attributes for clustering (Japan data): ["govt_debt_j", "interest_r_j", "gdp_pc_j"]. The analysis and values are recorded in the Figure 15.

Observations

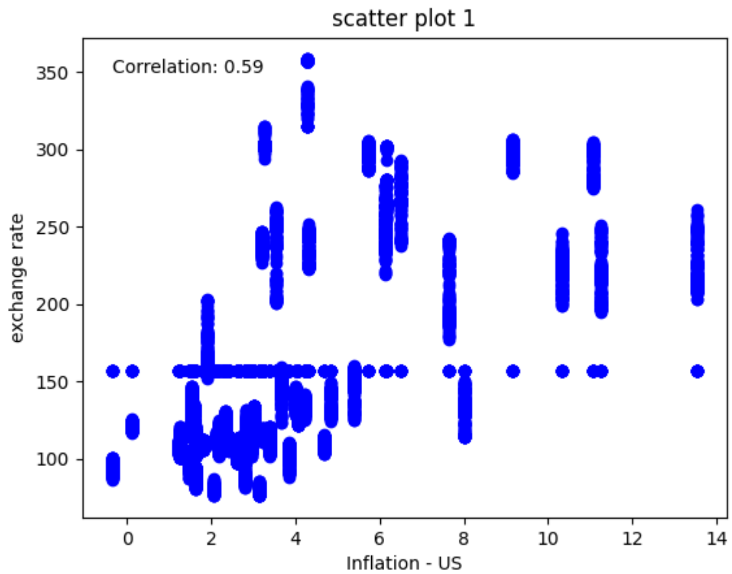


Figure 12: Scatter Plot (Exchange Rate Vs Inflation US)

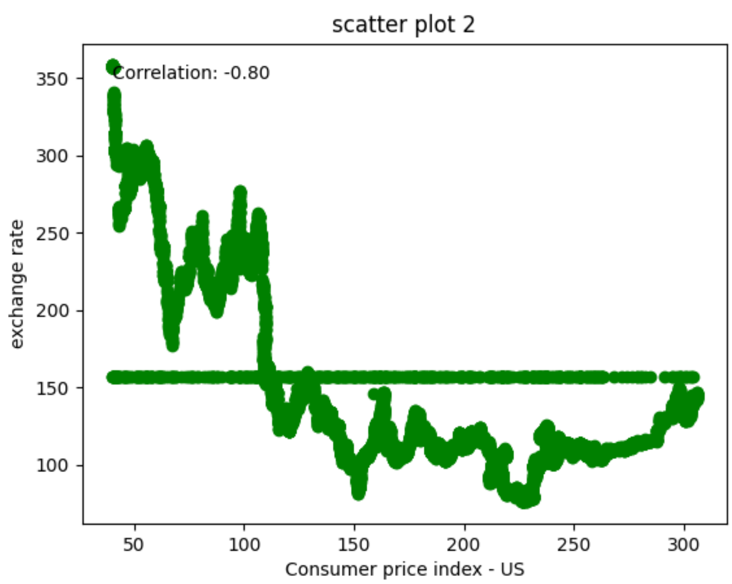


Figure 13: Scatter Plot ((Exchange Rate Vs CPI US))

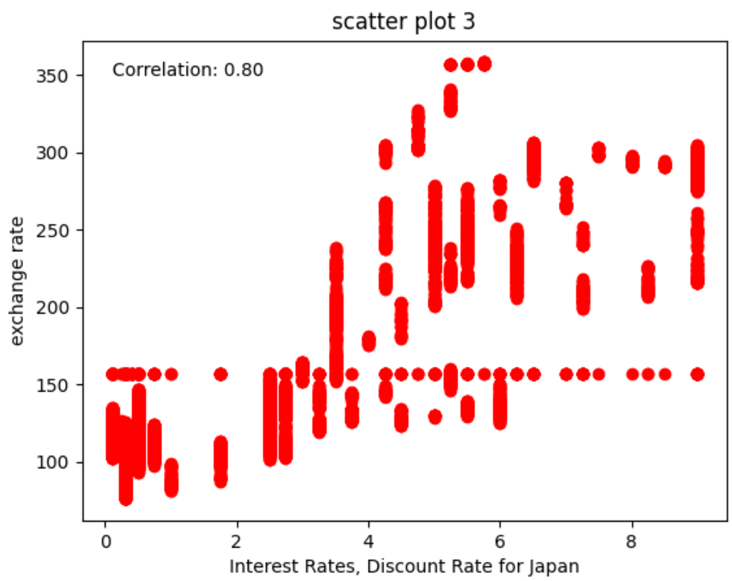


Figure 14: Scatter Plot (Exchange Rate Vs Interest Rate JP)

Experiments

No. of clusters	Silhouette score	Comments
2	0.7417698257537619	Good score
3	0.6224585621404772	Decrease in the silhouette score
4	0.6191595738761194	Increasing the clustering did not increase the score. No net gain.

Figure 15: K-Means Experiments

- Our Clusters improve as we reduce the number of clusters which are supported with a better silhouette score.
- The cluster sizes are not uniform, or close to being uniform.
- Since 2 cluster analysis was selected, the separation of the clusters can be seen but they are not very far apart from each other, but it is clear that there is a distinction and no overlap between any points. This indicates that our clusters are acceptable.
- Our clusters also have an imbalance in their sizes, which is indicative that there might still be some outliers.

DBSCAN

Attributes for clustering (Japan data): [”govt_debt_j”, ”interest_r_j”, ”gdp_pc_j”]. The analysis and values are recorded in the Figure 16.

EPS	Min points	Silhouette score	Comments
0.3	30	0.6308242317399431	There was an uneven distribution on size of clusters
0.5	30	0.6132253763905177	The score reduced as compared to the previous row, *so we tried increasing the min-points*
0.5	50	0.6437592320750956	the clusters looked a bit more uniform in terms of cluster sizing
0.5	80	0.5821840238795151	Increasing the min_points more after this simply resulted in a reduced silhouette score. *So we should try increasing eps*
1	80	0.6830564982547696	The best score till now, with the an increasing uniform distribution of clusters
Increasing the EPS values below :: which resulted in increasing silhouette scores			
3	80	0.7147826456738737	-
4	80	0.7480395151222803	-
7	80	0.8378521972173917	
12	80	0.914743529045488	cluster sizes got more uniform and there was one cluster which is considerably larger than the rest
Tried reduced the min_points and kept the same eps value (i.e 12)			
12	70	0.9360809529016481	Increase in silhouette score and better cluster balance
12	60	0.9412248332460468	Decided to stop here, because the clusters look more balanced in the no. of data points in each cluster and the silhouette score also looked good, but it is important to note that the density of clusters does not matter much and db_scan does a good job of handling it, so the density of clusters is not something that we need to worry)

Figure 16: DBSCAN Experiments

Table 2: Hierarchical Cluster Analysis

No. of Cluster	Score	Remark
2	0.74	Good Score. Also from other algorithms
4	0.60	Increasing the number of cluster did not increase the score
5	0.57	Score decreasing
6	0.58	Cannot increase score with increase in number of clusters

GMM

Attributes for clustering (Japan data): ["govt_debt_j", "interest_r_j", "gdp_pc_j"]. The analysis and values are recorded in the Figure 17. In GMM, the clusters might overlap which is fine.

No. of clusters	Silhouette score	Comments
3	0.488496250464599	Bad score
5	0.5628114368320333	the score improves with increase in no.of clusters and this method uses a soft clustering which helps us more as compared to k-means clustering)
6	0.5769249322147987	-
7	0.5376508359460689	Increasing the number of clusters is hurting us, and we did not find a significant increase in the silhouette scores (k-means had better scores comparatively when cluster = 2), so we should try GMM with 2 clusters
2	0.7358940532062618	Performs almost like k-means clustering, but only advantage here is that the covariance can be analyzed to check the different shapes of clusters which k-means cannot handle

Figure 17: GMM Experiments

Hierarchical

The attributes are same across all the algorithms to give an uniform analysis. The analysis is recorded in Table 2. We can observe that like other algorithms when the number of clusters is 2, we get the optimal score. The clusters can also be seen in Figure 18

Key Takeaway - We can improve our silhouette score, by searching for cluster imbalances, such as a cluster being too small and if it has outliers and then we can take this data out in order to improve the quality of our data.

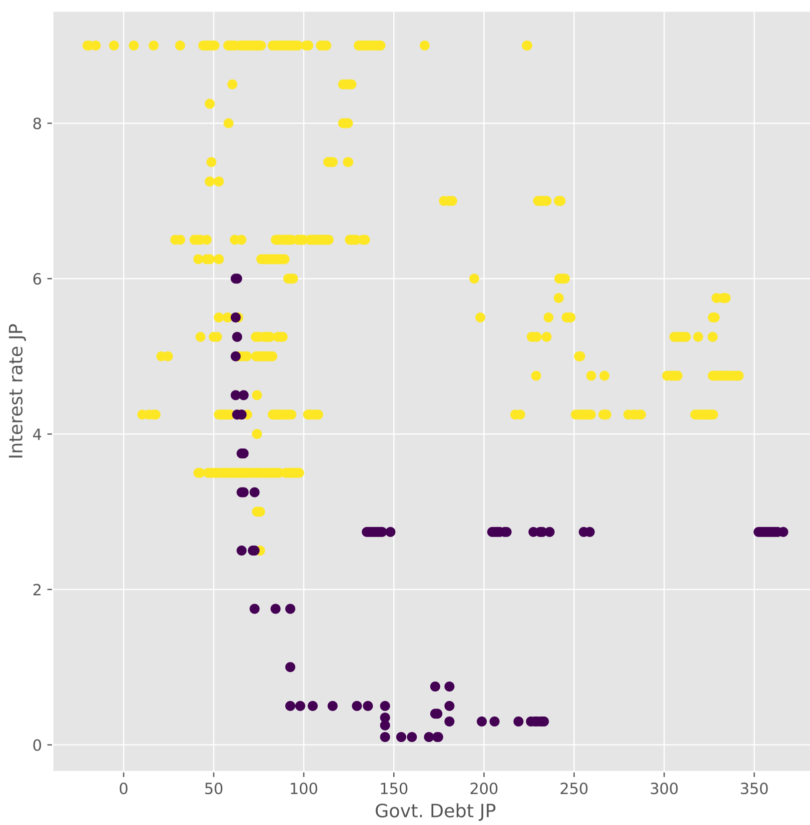


Figure 18: Hierarchical Clusters (2)

Sentiment Analysis

About the Data

For text, we are working with news data. We want to study the effects of real world events on the financial values that we analyze. We have collected news data from 2010 to 2023 from a wide variety of sources. Firstly, we have generic world news which covers occurrences of major events throughout the world. Then we further zoom in on the two countries, the United States and Japan. We further focus and download financial and business news relating to both the countries. Within financial news, we have 100 top articles for each year from 2010 to 2023.

We do sentiment analysis of the headlines of these news articles. We try to predict whether the headlines is "positive" - it's favorable news and is likely to be beneficial for the economy, "neutral", or "negative" - unfavorable and degrading towards the economy. We do it to capture the dependency of the financial values on real world events and to know whether the news is favorable to it or not. We plan to capture this dependency in future projects.

Text Preprocessing

Since we are working with news headlines, basic preprocessing was needed. We removed the source of the news article from the headline, removed special characters and converted the text to lower case.

Sentiment Analysis

We use *AFINN* and *VADER* as two comparative approaches for sentiment analysis. To test these approaches, we manually label 20 samples. Two human labelers sat together and labelled the first 20 samples. We verified our labels using a chatbot online. These labels are compared with the predictions of both these algorithms and their accuracy is obtained. We see that AFINN performs better with an accuracy of 0.8, while VADER performs with an accuracy of 0.75. We see a lot of mis-clasfication between "neutral" and "positive"/"negative", given the understandable ambiguity.